Problem 2(ii): From QED to QCD

With Explicit Color Vectors

PHYS 7502 — Particles and Nuclei 2

1 Question and Deliverables

Problem 2(ii)

Convert the Mandelstam variable amplitude for $e^-\mu^- \to e^-\mu^-$ to the equivalent $qq' \to qq'$ parton–parton scattering via gluon exchange. Different quark flavors (no u-channel). Sum/average over colors.

Deliverables

- Build the QCD amplitude using Griffiths' explicit color vectors c_i.
- Show how $c^\dagger \lambda^a c$ reduces to the matrix element λ^a_{ji}.
- Square, spin-sum, color-sum/average step by step.
- Final result in Mandelstam variables.

2 The Color Vectors

What Are the Color Vectors?

Every quark carries a color state, represented by a 3-component column vector in color space:

$$c_r = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad c_g = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad c_b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{1}$$

These are *not* Dirac spinors. They live in a completely separate 3-dimensional "color space." A quark's full state is really a tensor product: $u(p) \otimes c$, meaning a 4-component Dirac spinor times a 3-component color vector. But since the γ-matrices never touch the color vector and the λ^a-matrices never touch the spinor, the two factors always separate out, and we can handle them independently.

Analogy: Spin $\otimes$ Position in Non-Relativistic QM

This is just like writing a hydrogen atom state as $\psi(\mathbf{r})\,|\uparrow\rangle$. The spatial wavefunction and the spinor live in different spaces. When you compute $\langle H \rangle$, you do the spatial integral and the spin inner product separately. Here, the Dirac spinor is like $\psi(\mathbf{r})$ and the color vector is like $|\uparrow\rangle$.

3 Building the QCD Amplitude with Color Vectors

3.1 The QED Amplitude (for comparison)

From Griffiths 2nd ed. eq. 7.106:

$$\boxed{\mathcal{M}_{\text{QED}} = \frac{g_e^2}{t}\left[\bar{u}(p_3)\gamma^\mu u(p_1)\right]\left[\bar{u}(p_4)\gamma_\mu u(p_2)\right]} \tag{2}$$

No color vectors anywhere — the photon doesn't couple to color.

3.2 The QCD Amplitude (Griffiths' full notation)

From Griffiths 2nd ed. eq. 8.45–8.46. We label the four quarks:

Particle	Momentum	Spinor	Color vector
Incoming q	p_1	$u(p_1)$	c_1
Incoming q'	p_2	$u(p_2)$	c_2
Outgoing q	p_3	$\bar{u}(p_3)$	$c_3^\dagger$
Outgoing q'	p_4	$\bar{u}(p_4)$	$c_4^\dagger$

Now build the amplitude by following each fermion line backward:

Fermion Line 1 (Quark q): Full Notation

Follow backward from outgoing $q(p_3)$ to incoming $q(p_1)$:

Spinor part: $\bar{u}(p_3)\left[ig_e\gamma^\mu \to i\frac{g_s}{2}\gamma^\mu\right]u(p_1)$
Color part: $c_3^\dagger\left[\lambda^a\right]c_1$

Combined, the vertex contributes:

$$\left[\bar{u}(p_3)\,\gamma^\mu\,u(p_1)\right]\;\times\;\left[c_3^\dagger\,\lambda^a\,c_1\right]\;\times\;i\frac{g_s}{2} \tag{3}$$

The key point: the spinor bracket and the color bracket are **separate scalar factors**. The γ^μ acts on the 4-component spinors; the λ^a acts on the 3-component color vectors. They never mix.

Fermion Line 2 (Quark q'): Full Notation

Same procedure:

$$\left[\bar{u}(p_4)\,\gamma^\nu\,u(p_2)\right]\;\times\;\left[c_4^\dagger\,\lambda^b\,c_2\right]\;\times\;i\frac{g_s}{2} \tag{4}$$

Note: a different gluon color index b here, which will be contracted with a via the propagator.

Gluon Propagator

$$\frac{-ig_{\mu\nu}\,\delta^{ab}}{q^2} \tag{5}$$

The δ^{ab} forces $b = a$: the gluon emitted at vertex 1 must be the same gluon absorbed at vertex 2.

The $g_{\mu\nu}$ contracts the Lorentz indices $\mu \leftrightarrow \nu$ between the two spinor brackets.

Assemble the Full Amplitude

Multiplying everything together:

$$-i\mathcal{M} = \left[i\frac{g_s}{2}\,\bar{u}(3)\gamma^\mu u(1)\cdot c_3^\dagger \lambda^a c_1\right]\frac{-ig_{\mu\nu}\delta^{ab}}{t}\left[i\frac{g_s}{2}\,\bar{u}(4)\gamma^\nu u(2)\cdot c_4^\dagger \lambda^b c_2\right] \tag{6}$$

Handle the i's: $i\cdot(-i)\cdot i = -i$ ✓
Handle the δ^{ab}: Forces $b = a$, so $c_4^\dagger\lambda^b c_2 \to c_4^\dagger\lambda^a c_2$
Handle the $g_{\mu\nu}$: Lowers index, $\gamma^\nu \to \gamma_\mu$

Dividing by $-i$:

$$\boxed{\mathcal{M}_{\text{QCD}} = \frac{g_s^2}{4t}\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]\sum_{a=1}^{8}\left(c_3^\dagger\lambda^a c_1\right)\left(c_4^\dagger\lambda^a c_2\right)} \tag{7}$$

4 How $c^\dagger\lambda^a c$ Becomes λ^a_{ji}

Reducing the Color Bracket to a Matrix Element

The color bracket $c_3^\dagger\lambda^a c_1$ is a $\underbrace{(1\times 3)}_{c_3^\dagger}$ $\underbrace{(3\times 3)}_{\lambda^a}$ $\underbrace{(3\times 1)}_{c_1}$ matrix multiplication $\to$ a single number.

If quark 1 has color i (so c_1 has a 1 in position i and 0's elsewhere) and quark 3 has color j (so c_3 has a 1 in position j), then:

$$c_3^\dagger\,\lambda^a\,c_1 = (c_3)_j^*\,\lambda^a_{jm}\,(c_1)_m = \lambda^a_{ji} \tag{8}$$

It just **picks out the (j,i) matrix element** of λ^a. That's all it does.

Explicit example: Incoming red ($c_1 = c_r$), outgoing green ($c_3 = c_g$), gluon $a = 1$:

$$c_g^\dagger\,\lambda^1\,c_r = \begin{pmatrix}0 & 1 & 0\end{pmatrix}\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}1\\0\\0\end{pmatrix}$$

$$= \begin{pmatrix}0 & 1 & 0\end{pmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix} = 1 = \lambda^1_{21} \quad ✓ \tag{9}$$

Another example: Incoming red, outgoing red, gluon $a = 1$:

$$c_r^\dagger\,\lambda^1\,c_r = \begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix} = 0 = \lambda^1_{11} \quad ✓ \tag{10}$$

(Gluon $a = 1$ can't mediate $r \to r$ because λ^1 has a zero in the $(1,1)$ position.)

So Why Does Griffiths Write $c^\dagger \lambda^a c$ Instead of Just λ^a_{ji}?

Two reasons:

1. It makes the amplitude look more like QED: "adjoint × matrix × vector" is the same sandwich structure as $\bar{u}\gamma^\mu u$, just in color space instead of spinor space.
2. It makes it clear *where* the color information sits in the diagram — attached to the external quark lines, just like the spinors.

But operationally, $c_3^\dagger \lambda^a c_1 = \lambda^a_{ji}$ always. They are literally the same number.

With this identification, eq. (**??**) becomes:

$$\mathcal{M}_{\text{QCD}} = \frac{g_s^2}{4t} \left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right] \sum_{a=1}^{8} \lambda^a_{ji}\, \lambda^a_{lk} \tag{11}$$

where i, j are the incoming/outgoing colors of quark 1 and k, l are the incoming/outgoing colors of quark 2.

5 Squaring the Amplitude

$$|\mathcal{M}|^2 = \frac{g_s^4}{16\,t^2} \underbrace{\left|[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\right|^2}_{\text{Dirac: same as QED}} \times \underbrace{\left|\sum_a (c_3^\dagger \lambda^a c_1)(c_4^\dagger \lambda^a c_2)\right|^2}_{\text{Color}} \tag{12}$$

The factorization is exact because the two brackets are just numbers multiplying each other.

5.1 The Dirac Part (Identical to QED)

Casimir's trick, traces, massless limit (same as before):

$$\sum_{\text{spins}} \left|[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\right|^2 = 8(s^2+u^2) \tag{13}$$

(From the trace calculation: $32 \times \frac{s^2+u^2}{4}$.)

5.2 The Color Part (with explicit c's)

The color factor squared is:

$$\left|\sum_a (c_3^\dagger \lambda^a c_1)(c_4^\dagger \lambda^a c_2)\right|^2 = \sum_{a,b} (c_3^\dagger \lambda^a c_1)(c_4^\dagger \lambda^a c_2)\; (c_3^\dagger \lambda^b c_1)^*(c_4^\dagger \lambda^b c_2)^* \tag{14}$$

Complex Conjugate of a Color Bracket

Since $c^\dagger \lambda^a c$ is a (1×1) number, its complex conjugate is:

$$(c_3^\dagger \lambda^a c_1)^* = c_1^\dagger \lambda^{a\dagger} c_3 = c_1^\dagger \lambda^a c_3 \tag{15}$$

(using $\lambda^{a\dagger} = \lambda^a$, i.e. Hermiticity). In index notation:

$$(\lambda^a_{ji})^* = \lambda^a_{ij} \tag{16}$$

So conjugation just **swaps the two color indices**: incoming $\leftrightarrow$ outgoing.

The color factor squared becomes:

$$\sum_{a,b} \underbrace{(c_3^\dagger \lambda^a c_1)}_{=\lambda^a_{ji}} \underbrace{(c_1^\dagger \lambda^b c_3)}_{=\lambda^b_{ij}} \underbrace{(c_4^\dagger \lambda^a c_2)}_{=\lambda^a_{lk}} \underbrace{(c_2^\dagger \lambda^b c_4)}_{=\lambda^b_{kl}} \tag{17}$$

6 Color Sum/Average

6.1 What "Sum/Average" Means Operationally

We now **sum** over all possible final-state colors (c_3 and c_4 each range over r, g, b) and **average** over initial-state colors (c_1 and c_2 each range over r, g, b, divide by $3 \times 3 = 9$).

Summing Over a Color Vector

When we sum c_3 over all three color states r, g, b:

$$\sum_{c_3 \in \{r,g,b\}} (c_3^\dagger \lambda^a c_1)(c_1^\dagger \lambda^b c_3) = \sum_{j=1}^{3} \lambda^a_{ji}\, \lambda^b_{ij} \tag{18}$$

This is **summing over the index** j that c_3 controls. What's left? A sum over the repeated index j with a free index i — but we're also summing over c_1 (index i), so:

$$\sum_{c_1} \sum_{c_3} (c_3^\dagger \lambda^a c_1)(c_1^\dagger \lambda^b c_3) = \sum_{i=1}^{3} \sum_{j=1}^{3} \lambda^a_{ji}\, \lambda^b_{ij} = \sum_j (\lambda^a \lambda^b)_{jj} = \mathrm{Tr}(\lambda^a \lambda^b) \tag{19}$$

Summing over *both* color vectors for one fermion line gives a trace!

This is Just Like Casimir's Trick

In Casimir's trick, summing over spins using $\sum_s u\bar{u} = \not{p} + mc$ turns spinor products into *Dirac traces*.
Here, summing over colors using $\sum_j c_j c_j^\dagger = \mathbf{1}_{3\times 3}$ (the color completeness relation) turns color products into *color traces*. Same idea, different vector space.

Doing both fermion lines:

$$\sum_{\text{all colors}} |C_{\text{color}}|^2 = \sum_{a,b} \underbrace{\left(\sum_{c_1,c_3} (c_3^\dagger \lambda^a c_1)(c_1^\dagger \lambda^b c_3) \right)}_{\mathrm{Tr}(\lambda^a \lambda^b)} \underbrace{\left(\sum_{c_2,c_4} (c_4^\dagger \lambda^a c_2)(c_2^\dagger \lambda^b c_4) \right)}_{\mathrm{Tr}(\lambda^a \lambda^b)}$$

$$= \sum_{a,b} \left[\mathrm{Tr}(\lambda^a \lambda^b) \right]^2 \tag{20}$$

6.2 Applying $\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$

The Trace Orthogonality Identity

$$\mathrm{Tr}(\lambda^a\lambda^b) = 2\,\delta^{ab} \tag{21}$$

(Griffiths 2nd ed., eq. 8.52.) This says: if you pick two *different* Gell-Mann matrices and multiply them, the trace is zero. If you pick the *same* one twice, the trace is 2.

Proof sketch: The λ^a are traceless ($\mathrm{Tr}\lambda^a = 0$) and satisfy a specific normalization convention. The factor of 2 is a convention choice — it's what makes $T^a = \lambda^a/2$ normalized as $\mathrm{Tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, which is the standard physics normalization for SU(N) generators.

$$\sum_{a,b}\left[\mathrm{Tr}(\lambda^a\lambda^b)\right]^2 = \sum_{a,b}(2\delta^{ab})^2 = 4\sum_{a,b}\delta^{ab}\delta^{ab} = 4\sum_{a=1}^{8}1 = 32 \tag{22}$$

6.3 Include the Averaging Factor

Divide by 9 (average over 9 initial color states):

$$\langle|C|^2\rangle = \frac{1}{9}\times\frac{1}{16}\times 32 = \frac{32}{144} = \frac{2}{9} \tag{23}$$

(The $\frac{1}{16}$ is from $(g_s/2)^4 = g_s^4/16$, already sitting out front.)

Color-Averaged Color Factor

$$\boxed{\langle|C|^2\rangle = \frac{2}{9}} \tag{24}$$

7 Assembling the Final Result

$$\begin{aligned}\langle|\mathcal{M}|^2\rangle_{\mathrm{QCD}} &= \underbrace{\frac{1}{4}}_{\text{spin avg.}}\times\underbrace{\frac{1}{9}}_{\text{color avg.}}\times\underbrace{\frac{g_s^4}{16\,t^2}}_{\text{couplings}}\times\underbrace{8(s^2+u^2)}_{\text{Dirac traces}}\times\underbrace{32}_{\text{color sum}}\\ &= \frac{g_s^4(s^2+u^2)}{t^2}\times\frac{1}{4}\cdot\frac{1}{9}\cdot\frac{1}{16}\cdot 8\cdot 32\\ &= \frac{g_s^4(s^2+u^2)}{t^2}\times\frac{256}{576} = \frac{g_s^4(s^2+u^2)}{t^2}\times\frac{4}{9}\end{aligned} \tag{25}$$

Final Result

$$\boxed{\langle|\mathcal{M}|^2\rangle_{qq'\to qq'} = \frac{4}{9}\,g_s^4\,\frac{s^2+u^2}{t^2}} \tag{26}$$

For comparison:

$$\langle|\mathcal{M}|^2\rangle_{\text{QED}} = 2\,g_e^4\,\frac{s^2+u^2}{t^2}$$

Ratio:

$$\frac{\langle|\mathcal{M}|^2\rangle_{\text{QCD}}}{\langle|\mathcal{M}|^2\rangle_{\text{QED}}} = \frac{2}{9}\,\frac{g_s^4}{g_e^4}$$

8 Complete Factor-Tracking Table

Where Every Factor Lives

Factor	**QED**	**QCD**	**Origin**
Vertex coupling	g_e^2	$g_s^2/4$	$(g_s/2)^2$ from two vertices
Propagator	$1/t$	$1/t$	Same for γ and g
Spinor bracket 1	$\bar{u}(3)\gamma^\mu u(1)$	$\bar{u}(3)\gamma^\mu u(1)$	Identical
Spinor bracket 2	$\bar{u}(4)\gamma_\mu u(2)$	$\bar{u}(4)\gamma_\mu u(2)$	Identical
Color bracket 1	—	$c_3^\dagger \lambda^a c_1 = \lambda^a_{ji}$	New in QCD
Color bracket 2	—	$c_4^\dagger \lambda^a c_2 = \lambda^a_{lk}$	New in QCD
After squaring, summing, and averaging:			
Spin sum/avg	$\frac{1}{4}\cdot 8(s^2+u^2)$	$\frac{1}{4}\cdot 8(s^2+u^2)$	Casimir's trick
Color sum/avg	1	$\frac{1}{9}\cdot 32 = \frac{32}{9}$	Color Casimir
$\langle\|\mathcal{M}\|^2\rangle$	$\frac{2g_e^4(s^2+u^2)}{t^2}$	$\frac{4}{9}\frac{g_s^4(s^2+u^2)}{t^2}$	

9 Summary: The Path from $c^\dagger \lambda c$ to $\frac{2}{9}$

The Logic Chain

1. Each QCD vertex carries a color bracket $c^\dagger_{\text{out}} \lambda^a c_{\text{in}}$, which is a number equal to λ^a_{ji}.
2. The gluon propagator δ^{ab} forces both vertices to use the same gluon index a.
3. Squaring the amplitude produces $(c_3^\dagger \lambda^a c_1)(c_1^\dagger \lambda^b c_3)$ for each fermion line.

4. Summing c_1 and c_3 over r, g, b (sum final + sum initial before dividing) turns each fermion-line color product into $\mathrm{Tr}(\lambda^a \lambda^b)$.

5. The trace identity $\mathrm{Tr}(\lambda^a \lambda^b) = 2\delta^{ab}$ collapses the double gluon sum to $\sum_a 4 = 32$.

6. Dividing by 9 (average over initial colors) and 16 (from couplings) gives $32/144 = 2/9$.

7. This 2/9 multiplies the QED result with $g_e^4 \to g_s^4/4$, giving the final $\frac{4}{9} g_s^4$.